

Manual of ActFEM ver. 1.3

ACTIVE forward-inversion simulation code using FEM

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Contents

1. [Required environment](#)
2. [Coordinate system](#)
3. [Run samples for forward](#)
4. [Run samples for inversions](#)
5. [Source folder](#)
6. [Governing equation and discretization](#)
7. [File formats](#)
8. [Mesh generation tutorial](#)
9. [Generation of resistivity model for forward/inversion](#)

[Version note](#)

[References](#)

[Appendix A: Derivation of the discretized form of governing equation](#)

[Appendix B: Scaling of the governing equation for simulation](#)

1. Required environment

Intel compiler (ifx or ifort, mpiifort) for openMP and intel MPI.

There are serial modes only with openMP for forward modeling code.

However, there are no serial mode but only MPI parallel codes for inversions.

The required environments are as follows:

- Intel compiler (Linear system is solved with intel MKL library)

- gmsh

 - on Mac -> brew install gmsh

 - on Ubuntu -> sudo apt install gmsh

<only for viewing>

- GMT6

- on Mac -> brew install gmt

- on Ubuntu -> sudo apt install gmt

Gnuplot

- on Mac -> brew install gnuplot

- on Ubuntu -> sudo apt install gnuplot

gfortran is also applicable only for mesh generation (src/src_mesh) and post processing data (src/src_post)

2. Coordinate system

In ActFEM, commonly the following coordinate system is used:

X : East

Y : North

Z : Upward

3. Run samples for forward modeling

There is a forward sample folder. ActFEM/Fwd_test. Users can check how ActFEM calculates forward simulations: Active and MT.

<Step1: mesh generation (no topography) >

```
$cd ActFEM/Fwd_test/mesh_flat
```

```
$chmod +x meshgen.sh
```

```
$./meshgen.sh
```

meshgen.sh first compile the source in ActFEM/src/src_mesh with gfortran and run the program with gmsh. If you successfully installed gmsh, you can see the following output in the terminal.

```

====< nakadake3d.msh >====
# of nodes      (node) :    93381
# of elements   (nele) :   562869
# of Point      elements :      0
# of Line       elements :     17
# of Triangle   elements :      0
# of Tetrahedron elements :   562852
# of Traiangle group1 :      0
=====
### READ  nakadake3d.msh END!! ###

node-element connectivity end!
nline=    658152
### MKLINE END ###
### MKNGLINE END!! ###

```

Then please check the mesh by gmsh viewer:

\$gmsh nakadake3d.msh

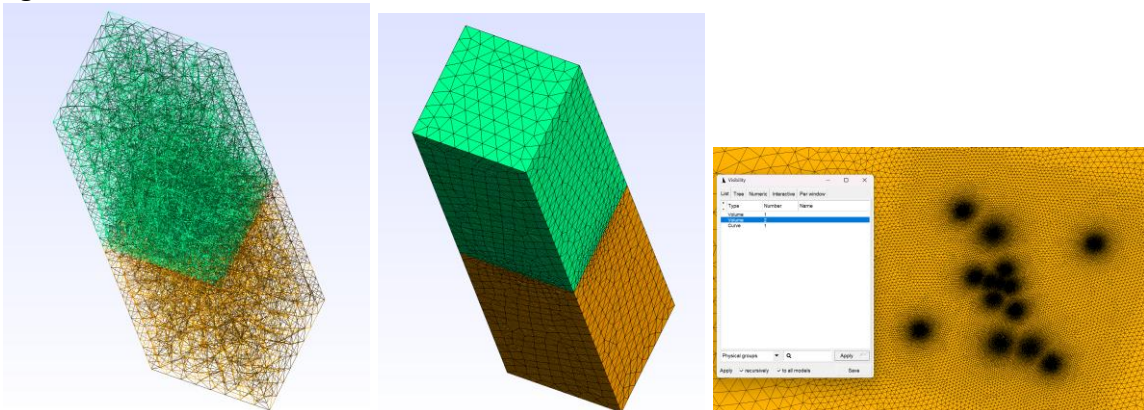


Figure. Tools>Options>Mesh>check 3D element faces shows the center panel.
 “Tools>Visibility>select Volume 2 > apply” shows only land elements, where you can check high resolution area for site locations.

<Step 2: Prepare analytical solution for ACTIVE >

For evaluating forward calculations, you can prepare analytical solution before forward modelling.

```
$cd ActFEM/Fwd_test/active_ana/
```

```
$chmod +x run.sh
```

```
$run.sh
```

As default, run.sh run with ifx to compile source programs in src/ana. If run.sh successfully finish, you can see:

```

<Input base depth of each layer up to nlayer-1 [km]
Sigma depth      1.000000[km]
<Input subsurface conductivity [S/m] for each layer>
  1sigma_land=    0.010000 [S/m]
  2sigma_land=    0.010000 [S/m]
### READ FORWARD PARAMETERS END!! ###
nfreq            5
nsr              2
nobs            4
### ALLOCATERESP END!! ###
      1 freq      1.000000000000000 [Hz]
      2 freq      3.000000000000000 [Hz]
      3 freq     11.000000000000000 [Hz]
      4 freq     33.000000000000000 [Hz]
      5 freq     99.000000000000000 [Hz]
### OUTOBSFILESFWD END!! ###

real    0m2.883s
user    0m2.856s
sys      0m0.012s

```

<Step 3: active forward modelling >

After generating mesh, you can run forward modelling for ACTIVE and MT.

```

$cd ActFEM/Fwd_test/active
$chmod +x run.sh
$./run.sh

```

If the active forward modelling finishes successfully, you can see:

```

### GENBCCSEM END!! ###
### SET_BC_ICCG END!! ###
### forward_bxyz END!! ### ip= 0 freq= 33.00000 Hz
### CALOBSEBCOMP END!! ###
< frequency = 99.000000 Hz >
nsr= 2 ip 0
### GENMAT END !!!!!
### GENBCCSEM END!! ###
### SET_BC_ICCG END!! ###
### forward_bxyz END!! ### ip= 0 freq= 99.00000 Hz
### CALOBSEBCOMP END!! ###
### OUTOBSFILESFWD END!! ###
### ebfem_bxyz END!! ### Time = 0h 5m 29.805s

real    5m29.928s
user    26m5.460s
sys      0m38.715s

```

<Step 4: compare the active forward results with analytical solution >

After you finish active modeling, you can check the accuracy of forward simulation with analytical solution in Fwd_test/active_ana/result.

In Fwd_test/active/

```

$chmod +x compana.sh
$./compana.sh

```

If the following panels appear, both active modelling and analytical solution calculations succeeded.

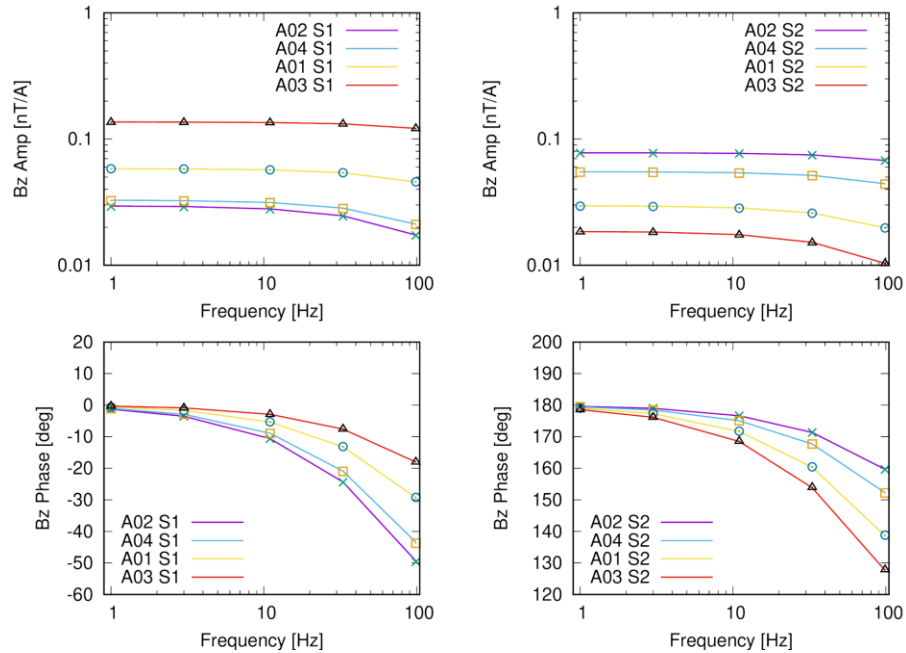


Figure. The symbols are the calculated ACTIVE response in Fwd_test/active/, while the lines are the analytical solutions calculated in Fwd_test/active_ana/. (Top: amplitude [nT/A]; Bottom: phase [deg])

<Step 5: MT forward calculation (without MPI) with homogeneous resistivity >

For MT calculations, there are serial (openMP) mode and Hybrid (openMP + MPI) mode. First, you can check serial model.

```
$cd ActFEM/Fwd_test/MTfwd_test
$chmod +x run.sh
$./run_homo.sh
```

If the calculation is successful, you can see the output:

```
### CALRESPTIP END !! ### ip = 0
Freq 100.000000000000 Hz
nline 658152
### GENMAT_MT END!! ### Time= 0.03486 [min]
### SET_BC_3DMT END!! ###
### forward_3DMT END!! ### ip= 0 freq=100.0000 [Hz] Time= 1.307[min]
### CALRESPMT END!! ###

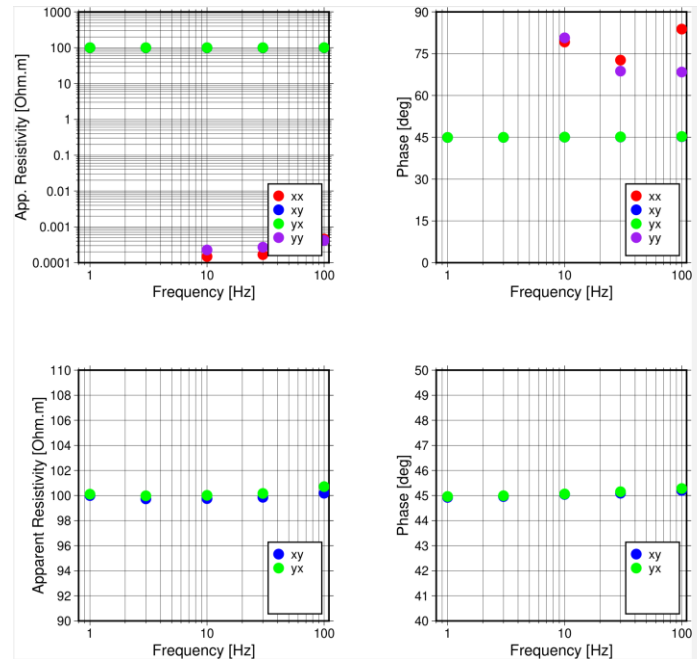
### CALRESPTIP END !! ### ip = 0
### OUTOBSRESPMT END!! ###
### OUTOBSRESP_TIP END!! ###

real 6m12.825s
user 28m53.087s
sys 0m39.076s
```

<Step 6: Check the results of MT forward calculation accuracy >

After finishing the calculations, please see how MT simulation finished.

```
$chmod +x plot_rho.sh
$./plot_rho.sh
```



As seen in the panels, MT simulation with ActFEM succeeded in generating 100 Ohm.m apparent resistivity and 45 degrees phase over the frequency ranging from 1 to 100 Hz with less than 1% numerical error.

<Step 7: MT forward calculation (without MPI) with lateral inhomogeneous structure>

MT forward calculation can calculate tipper as well. You can check how it works with laterally inhomogeneous structure. First it is need to generate the inhomogeneous structure. In ActFEM/Fwd_test/MTfwd_test/

```
$chmod +x change_model2cond.sh
```

```
./change_model2cond.sh
```

The output is like the following, resulting in **cond_lateral.msh**.

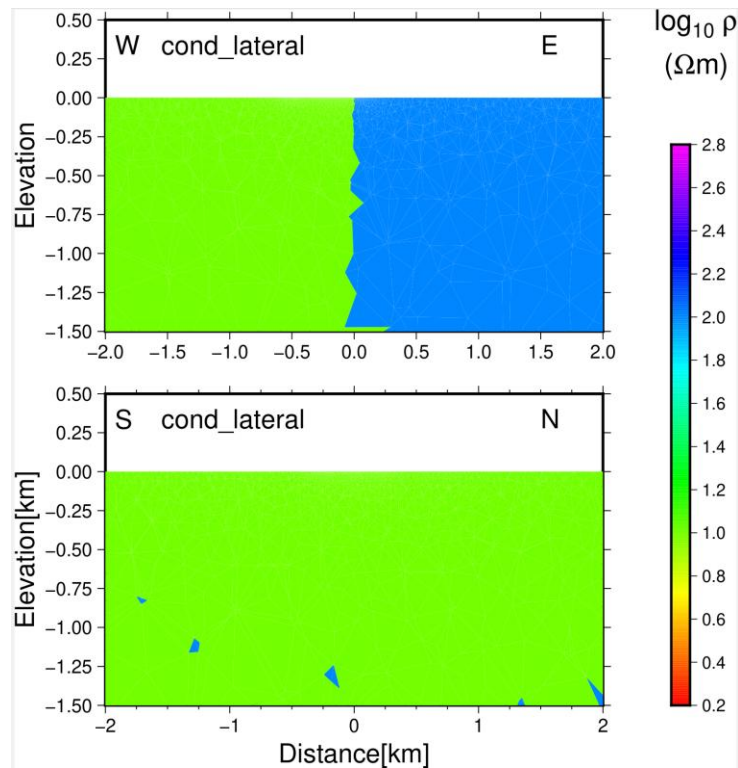
```
# of Tetrahedron elements :      0
# of Traiangle group1      :    71858
=====
### READ ../mesh_flat/nakadake2dz.msh END!! ###

ntri      71858
### allocate_2Dgrid_list END!! ###
classify2grd end!!
### ASSIGNRHO END!! ###
nphys1=    289235
nphys2=    273617
### OUTCOND END!! ###
```

To check the generated resistivity structure, type the below command:

```
$ chmod +x view_slcie.sh
```

```
$ view_slcie.sh
```



As seen in the WE and SN cross sections, the generated resistivity structure is lateral jump in resistivity at $x=0$ from 10 Ohm.m on the western side to 100 Ohm.m on the eastern side.

Then run the MT forward calculation with this structure by

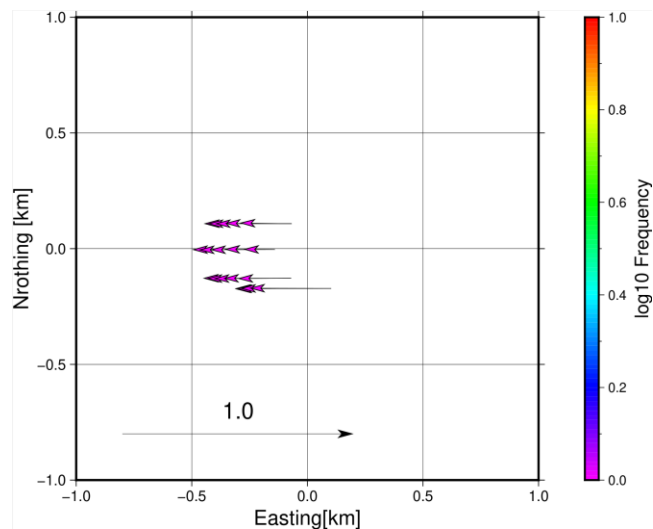
```
$ chmod +x run_lateral.sh
```

```
$ ./run_lateral.sh
```

After finishing the calculation, you can see the tipper by

```
$ chmod +x plot_tip.sh
```

```
$ ./plot_tip.sh
```



As seen in the panel, the real part of induction vector points west due to lower resistivity on the western side.

<Step 8: MT forward simulation with MPI + open MP >

Only MT forward simulation is now parallelized with MPI. You can do the same calculation as Fwd_test/MTfwd_test in Fwd_test/MTfwd_test_mpi/ with MPI.

```
$ cd Fwd_test/MTfwd_test_mpi/  
$ chmod +x run_homo.sh  
$ ./run_homo.sh
```

If the calculation is successful, the output is like:

```
### PREPOBSCOEFF END!! ###  
ip = 1/ 6 freq = 3.0000 [Hz] start!!  
### PREPOBSCOEFF END!! ###  
ip = 3/ 6 freq = 30.0000 [Hz] start!!  
### PREPOBSCOEFF END!! ###  
### ALLOCATERESP_MT END!! ###  
### ALLOCATERESP_MT END!! ###  
ip = 0/ 6 freq = 1.0000 [Hz] start!!  
### PREPOBSCOEFF END!! ###  
ip = 2/ 6 freq = 10.0000 [Hz] start!!  
### PREPOBSCOEFF END!! ###  
nline      658152  
nline      658152  
nline      658152  
nline      658152  
ip = 4/ 6 freq = 100.0000 [Hz] start!!  
nline      658152
```

Here you can see ip=1 (node#1) calculates freq=3.0Hz, while ip=3 (node#3) does freq=30Hz, etc. Each frequency calculation is assigned to each MPI process. Then the calculation is much faster than serial version. Please confirm the results in MTfwd_test/result_homo/ and MTfwd_test_mpi/result_homo/ are identical.

4. Run samples for inversions

In ActFEM, there 6 inversion samples: two ACTIVE, two MT, and two ACTIVE+MT joint inversion samples. To run inversions, prepare sample ACTIVE and MT data sets.

< Step 1: Prepare mesh >

Sample inversions are performed with real Aso topography. Let's generate mesh first.

```
$ cd ActFEM/Joint_test/mesh_joint  
$ chmod +x meshgen.sh  
$ ./meshgen.sh
```

After generating the mesh, please check how it looks by gmsh viewer.

```
$ gmth nakadake3d.msh
```

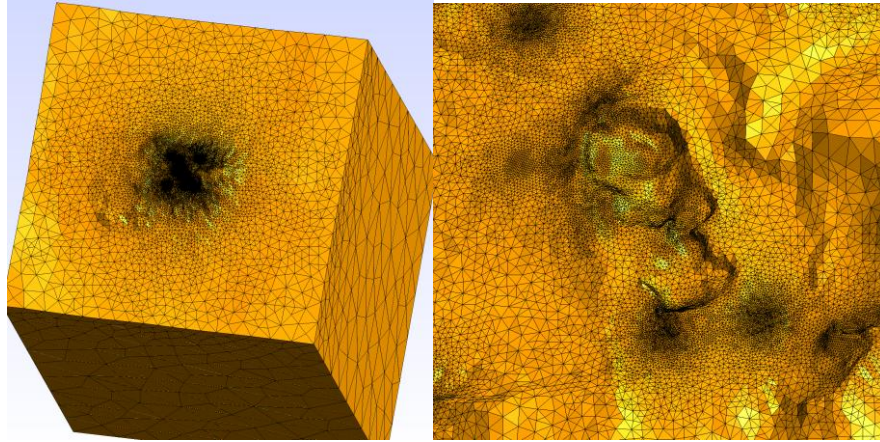



Figure. Generated tetrahedral mesh for Nakadake first crater of Aso volcano, Central Kyushu, Japan.

< Step 2: Prepare true resistivity structure >

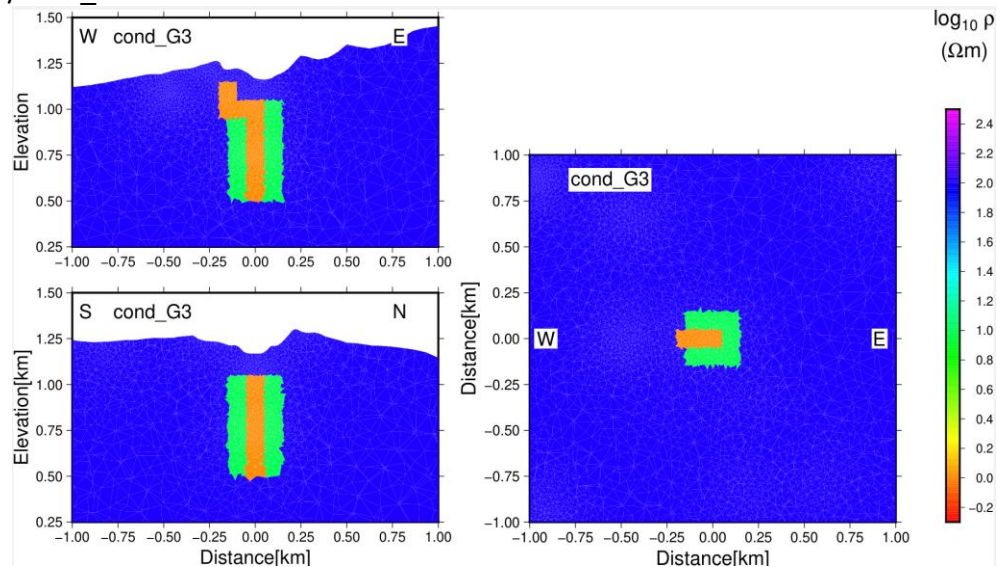
For inversion tutorials, we should assume some goal resistivity structure. ActFEM prepares the following goal resistivity structure in ActFEM/Joint_test/structure.

```
$ cd ActFEM/Joint_test/structure
$ ./cond_G3.sh
$ ./cond_homo.sh
```

If it is successful, **cond_G3.msh** and **cond_homo.msh** are generated. cond_G3.msh is the goal structure below and cond_homo.msh will be used as initial and reference homogeneous structure in the following inversions. **Note that cond_*.msh files are uniquely tied to original mesh_joint/nakadake3d.msh. If you update mesh_joint/nakadake3d.msh, please regenerate cond_G3.msh and cond_homo.msh.**

You can check how the anomaly of cond_G3.msh looks by

```
$ ./view_slice.sh
```



As seen in the panels, the anomaly is rectangular body extending just below the crater bottom.

< Step 3: Generation of ACTIVE sample data for inversion >

Now that we have true model, let's generate sample inversion data by forward modelling. Firstly, active data is prepared as follows:

```
$cd ActFEM/Joint_test/fwd_active/  
$ chmod +x run.sh  
$./run.sh
```

After finishing the forward calculation, prepare data with error in ActFEM/Joint_test/fwd_active:

```
$chmod + generr.sh  
$./generr.sh
```

As results, A0*_S?_amp.dat and A0*_S?_pha.dat are generated in Joint_test/fwd_active/data/. Here, A0* are site names written in active.ctl, while S? is the source name of electric dipole as written in active.ctl.

< Step4: Generation of MT sample data for inversion >

Secondly, MT data is prepared in ActFEM/Joint_test/fwd_3DMT/
\$cd ActFEM/Joint_test/fwd_3DMT/
\$chmod +x run.sh
\$./run.sh

The implementation generates **result/MT1_MT.dat** etc. The data with error are prepared by

```
$chmod +x generr.sh  
$./generr.sh
```

Then there are error files like **mt_err/MT1_MT_imp.err**, etc.

< Step 5: Run sample inversions!! >

Above, preparation for all the inversions are complete. The procedures for running inversion samples are the same. Run **run.sh**, and see the results by **plot_rms.sh** and **view_slice.sh**. For example, in Joint_test/inv_act/

```
$ cd ActFEM/Joint_test/inv_act/  
$./run.sh
```

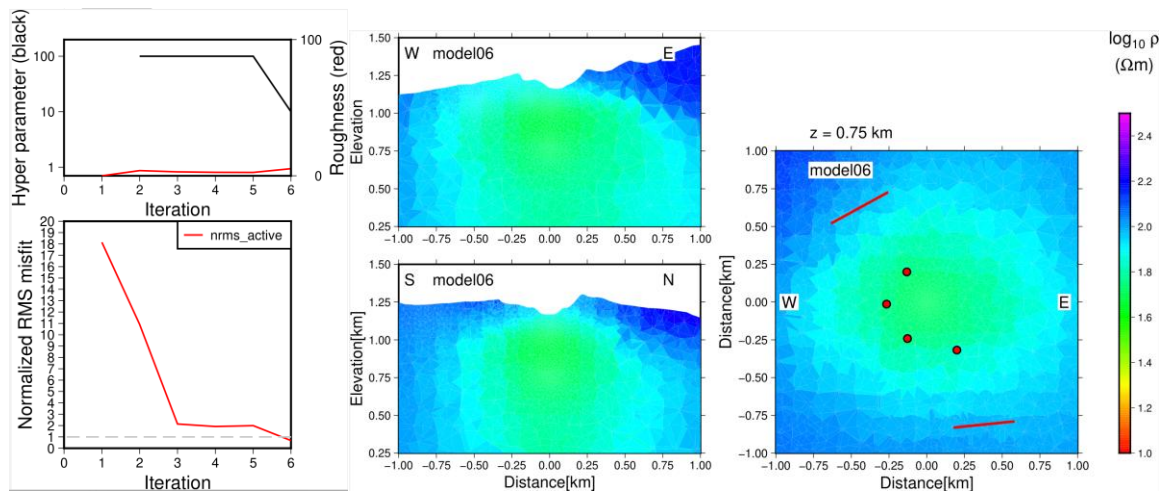
In run.sh, you should edit the number of OpenMP threads and MPI processes corresponding to your environment.

In the following, only the results are shown for obtained inversions results:

< Results of Joint_test/inv_act >

```
$plot_rms.sh  
$./view_slice.sh 06
```

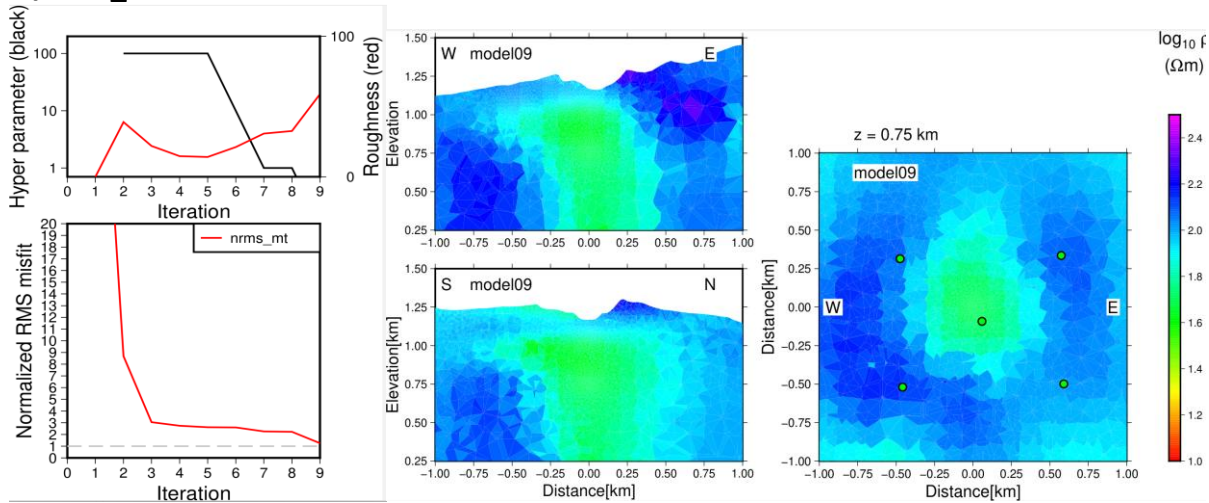
Here 06 means to show model06 which is converged model.



< Results of Joint_test/inv_mt/>

./plot_rms.sh

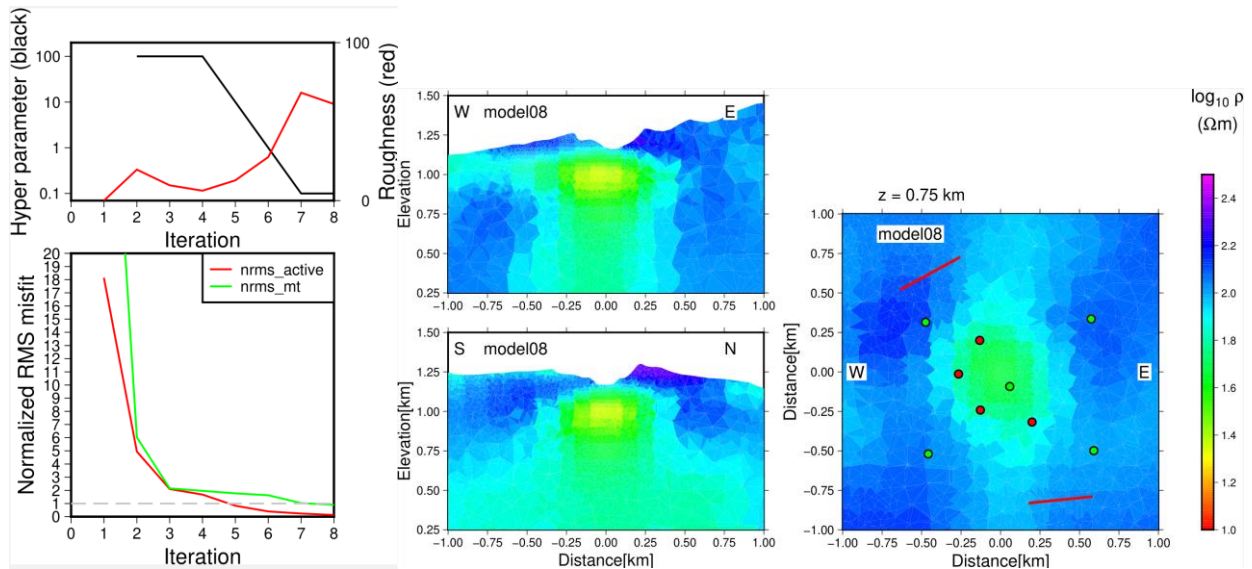
./view_slice.sh 09



< Result of Joint_test/inv_joint/ >

./plot_rms.sh

./view_slice.sh 08

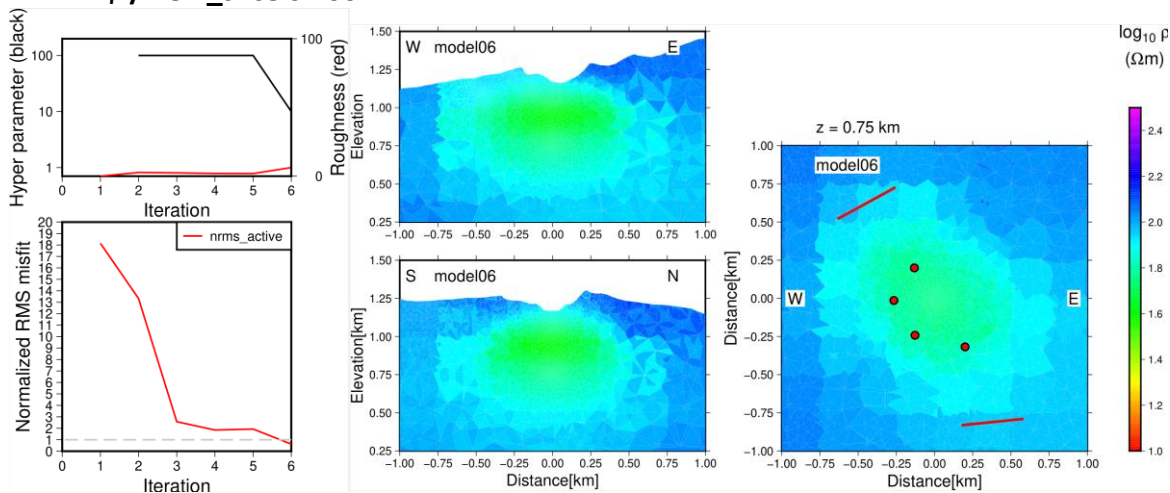


The following results are icombine = -1., which allows inversion to use inner outer model parametrization. The details are shown in the next section.

<Results of Joint_test/inv_act_icomb-1 >

\$plot_rms.sh

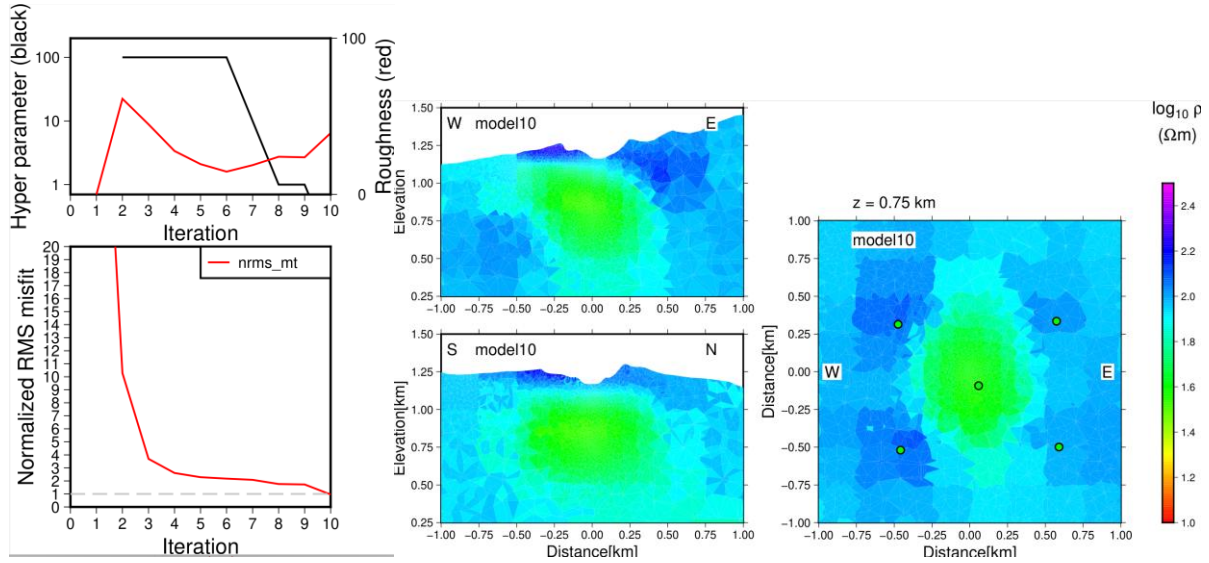
./view_slice.sh 06



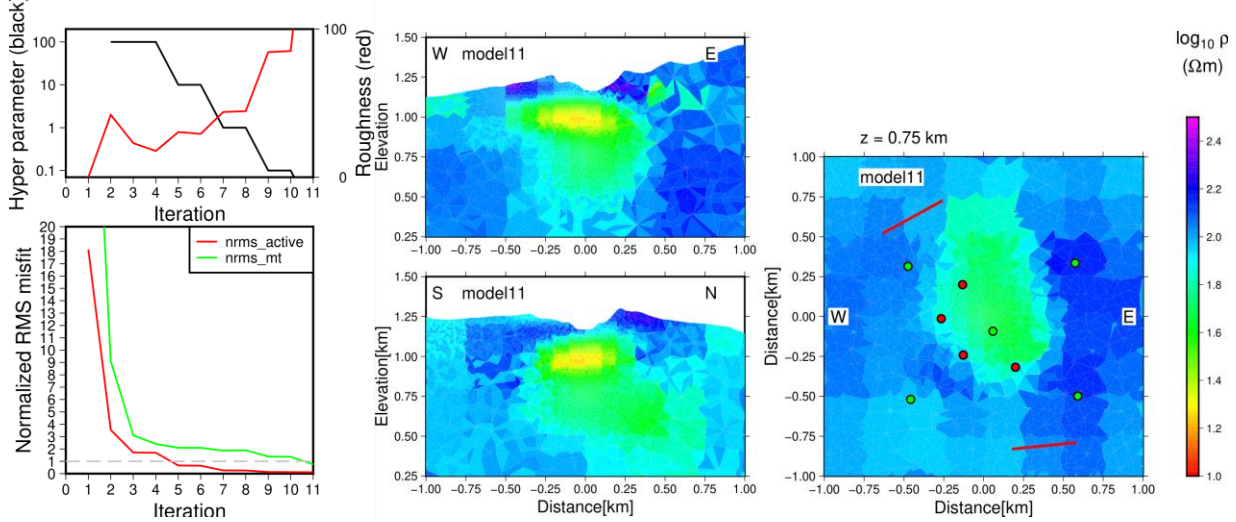
<Results of Joint_test/inv_mt_icomb-1 >

\$plot_rms.sh

./view_slice.sh 10



<Results of Joint_test/inv_joint_icomb-1 >



5. Source folder

ActFEMv1.0/src/solver_mpi/:

6. Governing equation and discretization

See Minami et al. (2018) for details. The induction equation in terms of the vector magnetic potential $\mathbf{A}$,

$$\nabla \times (\nabla \times \mathbf{A}) + i\omega\mu\hat{\sigma}\mathbf{A} = \mu\mathbf{i}_s$$

Is solved by the edge-based finite element method. $\hat{\sigma} = \sigma + i\omega\epsilon$, where ϵ is the electric permittivity. ω is the angular frequency, μ is the magnetic permeability, $\mathbf{i}_s$ is the source electric current density. Note that the equation is easily converted to the induction equation in terms of the electric field $\mathbf{E}$, by using the relationship of $\mathbf{E} = i\omega\mathbf{A}$,

$$\nabla \times \nabla \times \mathbf{E} + i\omega\mu\hat{\sigma}\mathbf{E} = i\omega\mu\mathbf{i}_s.$$

The magnetic field is obtained by $\mathbf{B} = \nabla \times \mathbf{A}$, after the equation is solved with respect to $\mathbf{A}$.

6. File formats

6.1 control files

Control file “nakaya.ctf” controls the forward calculation (green is explanation)

```
## lines starting with "##" work as lines for comments ! 2020.09.28
## this file is forward control file
##-----10!-----20!
itopofile 0 or 1 !1 (0:地形ファイルを使わずメッシュ生成、1:地形ファイルを使う。)
# of topofile !1 (使う地形ファイルの数。以下、狭い領域のファイルから、広い領域のファイルの順で、入力する。)
topofile !../topo/topo127_134_29_36.xyz (地形ファイル名)
lon lat shift !0.0 0.0 (lon latの実際の値からのshift値)
mesh file !../mesh_aso_A04/nakadake3d.msh (3-D mesh file used in the fwd calc.)
2d triangle z file !../mesh_aso_A04/nakadake2dz.msh (2-D mesh with topography info)
local line file !../mesh_aso_A04/lineinfo.dat (line information file)
output folder !./result_fwd/
header2d (a50) !nakadake2d
header3d (a50) !nakadake3d
# of frequency !2 (number of frequency for calculation)
Frequency [Hz] !1.d0 (first frequency)
Frequency [Hz] !3.d0 (second frequency)
##Frequency [Hz] !7.d0 (commented out line starting with "##")
##Frequency [Hz] !11.d0 (commented out line starting with "##")
##Frequency [Hz] !21.d0 (commented out line starting with "##")
##Frequency [Hz] !41.d0 (commented out line starting with "##")
##Frequency [Hz] !61.d0 (commented out line starting with "##")
##Frequency [Hz] !99.d0 (commented out line starting with "##")
west bound !-1.7 (mesh parameter, not used in forward calc.)
east bound !1.7 (mesh parameter, not used in forward calc.)
south bound !-1.5 (mesh parameter, not used in forward calc.)
north bound !1.5 (mesh parameter, not used in forward calc.)
lenout [km] !50.0 (mesh parameter, not used in forward calc.)
upz in [km] (>0) !1.3 (mesh parameter, not used in forward calc.)
downz in [km](<0) !-1.1 (mesh parameter, not used in forward calc.)
zmax [km] !50.0 (mesh parameter, not used in forward calc.)
zmin [km] !-50.0 (mesh parameter, not used in forward calc.)
sizein [km] !0.15 (mesh parameter, not used in forward calc.)
sizebo [km] !10.0 (mesh parameter, not used in forward calc.)
sigma_obs [km] !0.4 (mesh parameter, not used in forward calc.)
A_obs [km] !0.01 (mesh parameter, not used in forward calc.)
dlen_source [km] !0.1 (mesh parameter, not used in forward calc.)
sigma_src [km] !0.3 (mesh parameter, not used in forward calc.)
A_src [km] !0.005 (mesh parameter, not used in forward calc.)
# of observatory !4 (# of observation sites for calculation of magnetic/electric field)
lonlat(1),xyz (2) !1 (1 for lon lat input, 2 for xyz input)
UTM_ZONE !52S (UTM zone when lonlat input)
lonlatorigin !131.084782 32.884882 (lon lat origin of mesh)
1 Name !A02 (Site name 1)
1 xyz !131.083411 32.886706 -0.001 (Lon lat alt, alt is from ground surface)
2 Name !A04 (Site 2)
2 xyz,sigma,A[km] !131.081939 32.884808 -0.001 (Lon lat alt, alt is from ground surface)
3 Name !A01 (Site 3)
3 xyz,sigma,A[km] !131.083367 32.882725 -0.001 (Lon lat alt, alt is from ground surface)
4 Name !A03 (Site 4)
4 xyz,sigma,A[km] !131.086847 32.881981 -0.001 (Lon lat alt, alt is from ground surface)
ixyflag 0:no,1:surfv!0 (surface map flag)
# of sources !1 (number of electric dipole sources)
Source Name !S1 (name of the first source)
source start point !131.0784333 32.8908028 -0.001 (dipole starting point)
```

source end point !131.0814639 32.8912333 -0.001 (dipole ending point)
Electric current[A]! 1.0 (value of electric current through the electric dipole)
sigma_air [S/m] !1.e-8 (electric conductivity of the air)
condflag 0:home,1: !0 (subsurface conductivity flag, 0: homogeneous earth, 1 for file input)
nvolume !1 (when flag is 0,number of volume is specified here)
cond !0.01 (conductivity of the first volume under the ground surface)

7. Mesh generation tutorial

The mesh generation for FEM modelling is one of toughest works for forward/inversion preparation. Here, simple tutorial help the users to utilize the mesh generation codes in ActFEM forward/inversion. Please start the tutorial in the folder, [ActFEMv1.0/mesh_demo_Aso/](#).

In [ActFEM/mesh_demo_Aso/](#), users can two different 3-D mesh files using [meshgen.sh](#) and [meshgen_add.sh](#), where the only difference is the adopted control file, [aso.ctl](#) and [aso_add.ctl](#). In [ActFEM/mesh_demo_Aso/](#), you can just perform the following

```
$ ./meshgen.sh
```

or

```
$ ./meshgen_add.sh
```

Note that all the output file names by [meshgen.sh](#) and [meshgen_add.sh](#) are the same. Please conduct

```
$ ./clean.sh
```

in [ActFEM/mesh_demo_Aso/](#), if you want to delete all the output files generated by [meshgen.sh](#) or [meshgen_add.sh](#). The difference between the two scripts, i.e. the difference in two control files, is the number of receiver sites, where, only in [aso_add.ctl](#), receiver B02 is added, which results in the total number of receivers of 8 in [aso_add.ctl](#) rather than 7 in [aso.ctl](#). The difference is shown in the following figure:

36	# of observatory	!7	36	# of observatory	!8
37	lonlat(1),xyz (2)	!1	37	lonlat(1),xyz (2)	!1
38	UTM_ZONE	52S	38	UTM_ZONE	52S
39	lonlatorigin	!131.084782 32.884882	39	lonlatorigin	!131.084782 32.884882
40	1 Name	!A02	40	1 Name	!A02
41	1 xyz	!131.083411 32.886706 -0.001	41	1 xyz	!131.083411 32.886706 -0.001
42	2 Name	!A04	42	2 Name	!A04
43	2 xyz,sigma,A[km]	!131.081939 32.884808 -0.001	43	2 xyz,sigma,A[km]	!131.081939 32.884808 -0.001
44	3 Name	!A01	44	3 Name	!A01
45	3 xyz,sigma,A[km]	!131.083367 32.882725 -0.001	45	3 xyz,sigma,A[km]	!131.083367 32.882725 -0.001
46	4 Name	!A03	46	4 Name	!A03
47	4 xyz,sigma,A[km]	!131.086847 32.881981 -0.001	47	4 xyz,sigma,A[km]	!131.086847 32.881981 -0.001
48	5 Name	!DUM	48	5 Name	!B02
49	5 xyz,sigma,A[km]	!131.084782 32.884882 -0.100	49	5 xyz,sigma,A[km]	!131.087825 32.885672 -0.001
50	6 Name	!DM2	50	6 Name	!DUM
51	6 xyz,sigma,A[km]	!131.083335 32.884882 -0.100	51	6 xyz,sigma,A[km]	!131.084782 32.884882 -0.100
52	7 Name	!DM3	52	7 Name	!DM2
53	7 xyz,sigma,A[km]	!131.079672 32.884945 -0.100	53	7 xyz,sigma,A[km]	!131.083335 32.884882 -0.100
			54	8 Name	!DM3
			55	8 xyz,sigma,A[km]	!131.079672 32.884945 -0.100

Receivers set for mesh refinement in [aso.ctl](#) (Left), and in [aso_add.ctl](#) (Right). B02 is added in [aso_add.ctl](#).

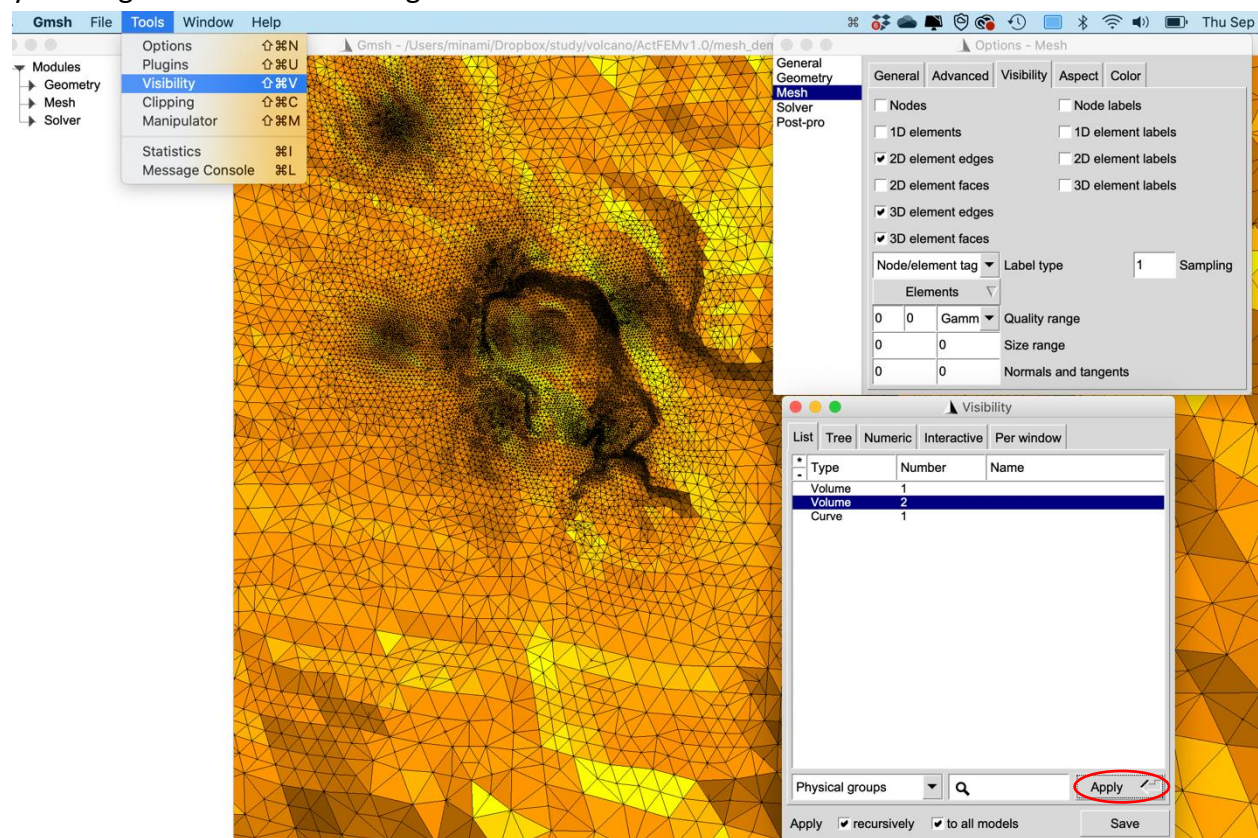
When you succeeded in running [meshgen.sh](#) or [meshgen_add.sh](#) and generating 3-D mesh for ActFEM forward/inversion simulation, you can get the following files:

```

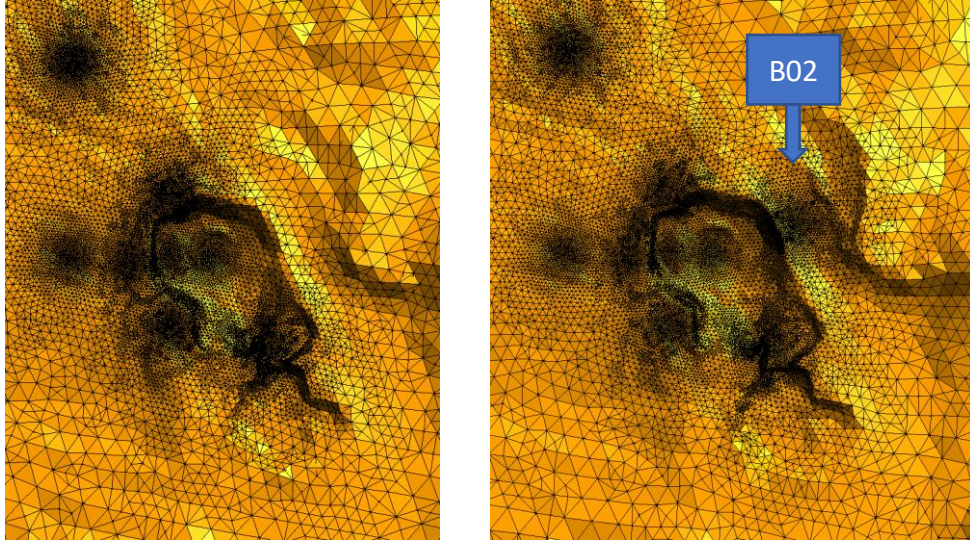
aso.ctl
aso_add.ctl
clean.sh
meshgen.sh
meshgen_add.sh
nakadake2d.geo
nakadake2d.msh
nakadake2d.pos
nakadake2d_lonlatz.msh
nakadake2d_z.msh
nakadake2dz.msh
nakadake3d.geo
nakadake3d.log
nakadake3d.msh
nakadake3d.pos

```

See meshgen.sh to check when each output files are generated. The final output 3-D tetrahedral mesh is **nakadake3d.msh**. You can check the detail of the generated 3-D tetrahedral mesh by gmsh GUI (Graphical User Interface). “Tools > Visibility > Selection of Volume 2 > Apply” shows you the ground surface of the generated 3-D mesh.



Then please check the following two pictures of ground surface of generated 3-D meshes. You can find the refined region corresponding to B02 in the 3-D mesh generated by **meshgen_add.sh** (i.e. **aso_add.ctl**). As the modification from **aso.ctl** to **aso_add.ctl**, the users can modify the resulting 3-D mesh.



Nakadake3d.msh generated by `meshgen.sh` (Left) and that generated by `meshgen_add.sh` (Right)

8. Generation of resistivity model for forward/inversion

For forward/inversion using ActFEM, users should prepare resistivity model for a given 3-D tetrahedral mesh. Here, simple tutorial help the users to build a resistivity model for a 3-D mesh generated in the previous section. Then please conduct `meshgen.sh` or `meshgen_add.sh` in `ActFEMv1.0/mesh_demo_Aso/` before trying the tutorials here. (See Section 7 for details.)

To try tutorials here, first move to the folder `ActFEMv1.0/mesh_demo_rec2tet/`, where the resistivity model is constructed based on `ActFEMv1.0/mesh_demo_Aso/nakadake3d.msh`. In `ActFEMv1.0/mesh_demo_rec2tet/`, run

```
$. /condmesh.sh
```

If you succeeded in running `condmesh.sh`, you can obtain the following files:

```
clean.sh      condmesh.sh
cond_init.ctl init.msh
cond_init.msh
```

`cond_init.ctl` is not an output file but the control file for running `condmesh.sh`. `cond_init.msh` is the file you can use in the forward / inversion in ActFEM. Generated `cond_init.msh` is the file used in the initial model in Minami et al. (2018, EPS) based on Kanda et al. (2018, JpGU). Note that you cannot view the `cond_init.msh` simply by `$gmsch cond_init.msh`, because `cond_init.msh` doesn't include the 3-D mesh information. `init.msh` is the file combining the `mesh_demo_rec2tet/cond_init.msh` and `mesh_demo_Aso/nakadake3d.msh`. The structure difference between `cond_init.msh` and `init.msh` are as follows:

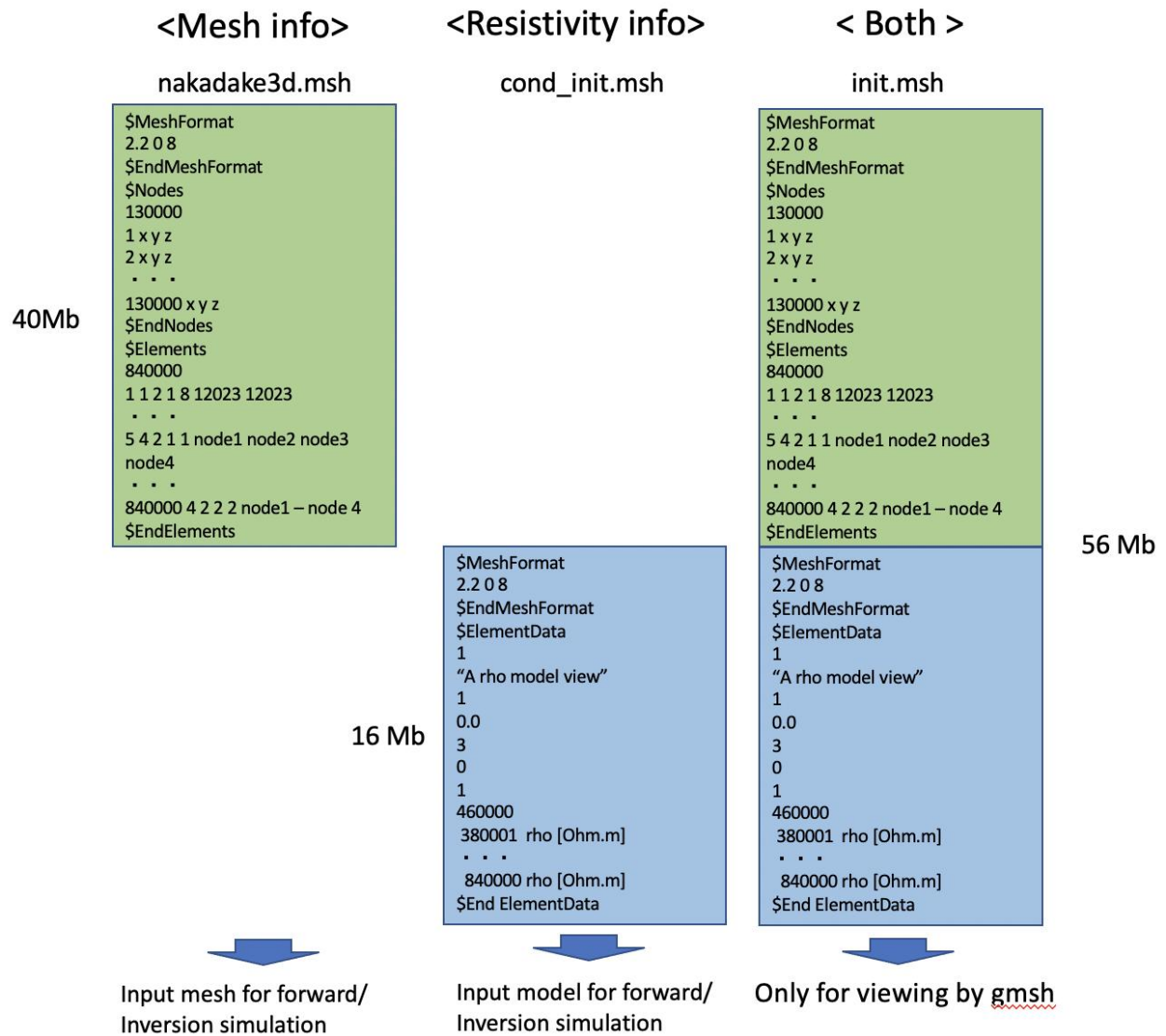


Figure Structure of nakadake3d.msh, cond_init.msh, and init.msh

Then you can see the generated resistivity model by

\$ gmsh init.msh

and set some viewer parameters as shown in the picture below.

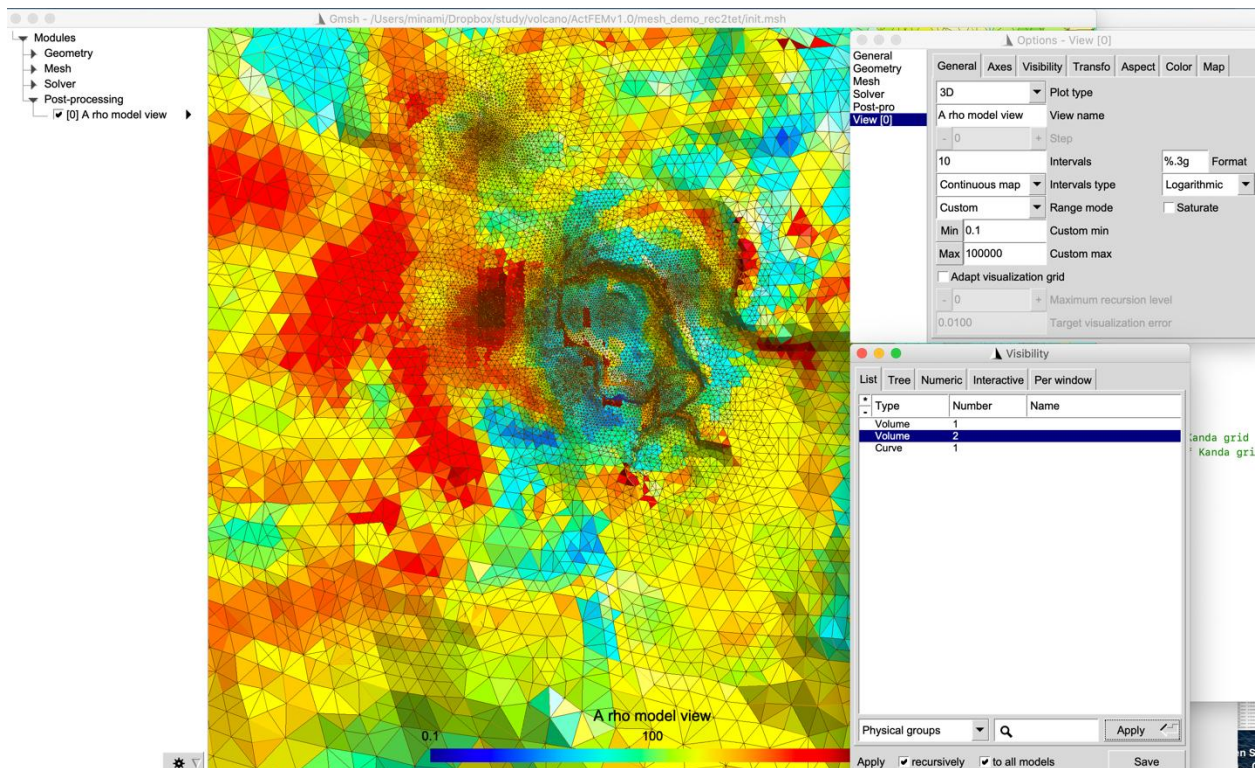


Figure View of init.msh by gmsh GUI. If users want to inverse the color scale, namely if you want to make high resistivity in cold color, click Map tab in Option Panel, and press “i”.

In the script `condmesh.sh`, `ActFEMv1.0/src/src_mesh/gridmdl2mesh.exe` is used to generate `cond_init.msh`. The schematic for what `condmesh.sh` does is in the following figure. Then please change `cond_init.ctl` for modification of output resistivity file.

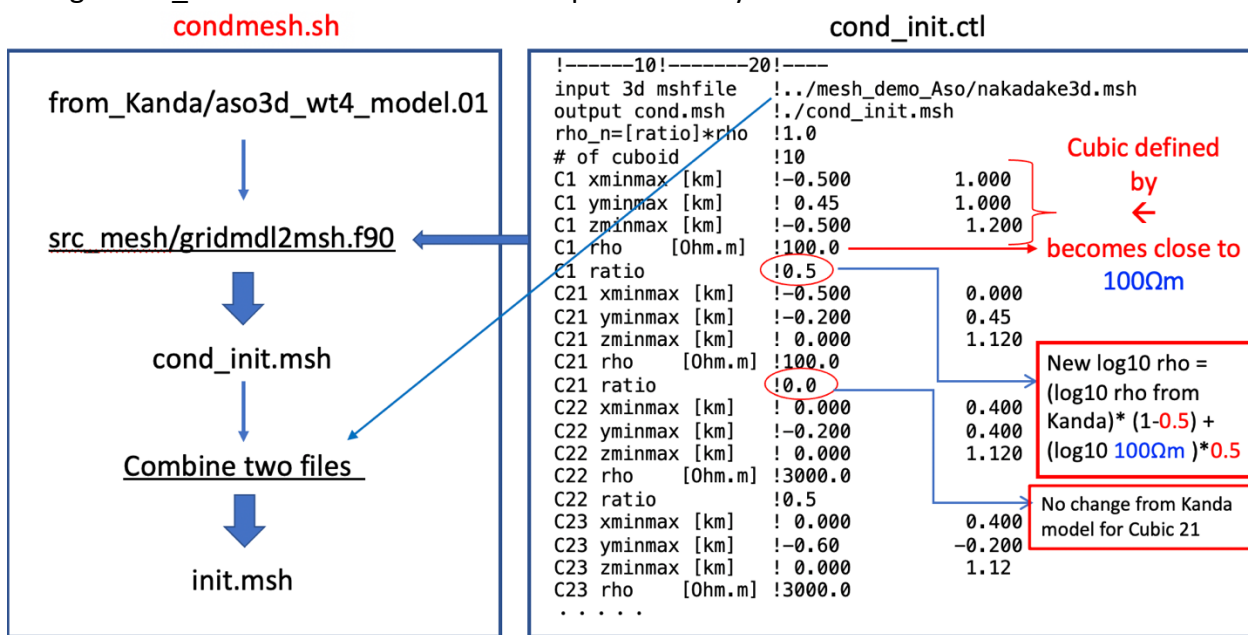


Figure Outline of condmesh.sh

Version Notes

Version 1.0

Latest date for update is Sep 30, 2021. Forward and inversion simulation can be available.

References

Minami, T., Utsugi, M., Utada, H., Kagiya, T., & Inoue, H. (2018). Temporal variation in the resistivity structure of the first Nakadake crater, Aso volcano, Japan, during the magmatic eruptions from November 2014 to May 2015, as inferred by the ACTIVE electromagnetic monitoring system. *Earth, Planets and Space*, 70(1), 1-10.

Appendix A: Derivation of the discretized form of governing equation

Given the governing equation,

$$\nabla \times \nabla \times \mathbf{A} + i\omega\mu\sigma\mathbf{A} = \mu\mathbf{i}_s,$$

in the finite element method with the Galerkin method, we numerically solve

$$\int_{\Omega} \mathbf{w} \cdot (\nabla \times \nabla \times \mathbf{A} + i\omega\mu\sigma\mathbf{A} - \mu\mathbf{i}_s) dV = 0, \dots (1)$$

where w is arbitrary test function, where the edge-basis function is adopted for $\mathbf{w}$ in the Galerkin method, and Ω is the computational domain.

By using the relationship $\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$, the first term in the left hand side of Eq. (1) is reduced to

$$\begin{aligned} \int_{\Omega} \mathbf{w} \cdot (\nabla \times \nabla \times \mathbf{A}) dV &= \int_{\Omega} \{ \nabla \cdot ((\nabla \times \mathbf{A}) \times \mathbf{w}) + (\nabla \times \mathbf{w}) \cdot (\nabla \times \mathbf{A}) \} dV \\ &= \int_{\Omega} \{ \nabla \cdot ((\nabla \times \mathbf{A}) \times \mathbf{w}) + (\nabla \times \mathbf{w}) \cdot (\nabla \times \mathbf{A}) \} dV \\ &= \int_{\partial\Omega} ((\nabla \times \mathbf{A}) \times \mathbf{w}) \cdot \mathbf{n} dS + \int_{\Omega} \{ (\nabla \times \mathbf{w}) \cdot (\nabla \times \mathbf{A}) \} dV \\ &= \int_{\partial\Omega} (\mathbf{B} \times \mathbf{w}) \cdot \mathbf{n} dS + \int_{\Omega} \{ (\nabla \times \mathbf{w}) \cdot (\nabla \times \mathbf{A}) \} dV \dots (2) \end{aligned}$$

Where $\mathbf{B} = \nabla \times \mathbf{A}$ represents the magnetic field and $\partial\Omega$ indicates the boundary surface of the computational domain, and $\mathbf{n}$ is a outward unit vector normal to the surface $\partial\Omega$. Taking into account that at $\partial\Omega$, we can take

$$\mathbf{n} \cdot (\mathbf{B} \times \mathbf{w}) = \mathbf{w} \cdot (\mathbf{n} \times \mathbf{B}) = 0,$$

where $\nabla \cdot (\mathbf{n} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{n}) - \mathbf{n} \cdot (\nabla \times \mathbf{B}) = -\mathbf{n} \cdot (\mu\mathbf{j}) = 0$ indicates there are no electric current penetrating the boundary surface. Then, Eq. (2) reduces to

$$\int_{\Omega} \mathbf{w} \cdot (\nabla \times \nabla \times \mathbf{A}) dV = \int_{\Omega} \{(\nabla \times \mathbf{w}) \cdot (\nabla \times \mathbf{A})\} dV \dots (3)$$

Substituting Eq. (3) into Eq. (1), we obtain

$$\int_{\Omega} \{(\nabla \times \mathbf{w}) \cdot (\nabla \times \mathbf{A})\} dV + \int_{\Omega} i\omega\mu\sigma \mathbf{w} \cdot \mathbf{A} dV = \mu \int_{\Omega} \mathbf{w} \cdot \mathbf{i}_s dV \dots (4)$$

In the finite element method, we divide the numerical domain into finite number of the tetrahedrons and calculate the volume integral for each tetrahedral element. In the remaining, we think only an volume integral for e-th element, Ω_e , then Eq. (4) is rewritten for e-th element as

$$\int_{\Omega_e} \{(\nabla \times \mathbf{w}_i) \cdot (\nabla \times \mathbf{A}_e)\} dV + \int_{\Omega_e} i\omega\mu\sigma \mathbf{w}_i \cdot \mathbf{A}_e dV = \mu \int_{\Omega_e} \mathbf{w}_i \cdot \mathbf{i}_{s_e} dV \dots (5)$$

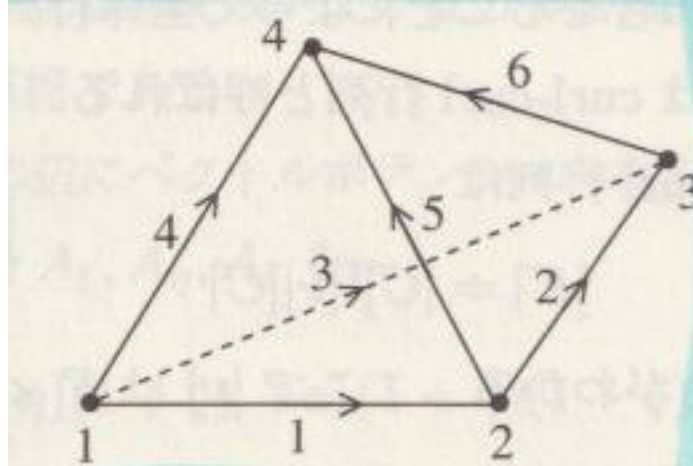
where w_i ($i = 1, \dots, 6$) is the edge-basis function for each edge of the e-th tetrahedron. We represent the vector potential in the e-th element as

$$\mathbf{A}_e = \sum_{j=1}^6 \mathbf{w}_j [A]_j \dots (6)$$

where

$$\mathbf{w}_j = \mathbf{w}_{kl} = \lambda_k \nabla \lambda_l - \lambda_l \nabla \lambda_k.$$

λ_k is the nodal basis function. The relationship between j and k,l are shown in the following figure and table.



i, j (edge id)	k	l	m	n
1	1	2	3	4
2	2	3	1	4
3	1	3	4	2
4	1	4	2	3
5	2	4	3	1
6	3	4	1	2

Table A1

Using Eq. (6), Eq. (5) can be reduced to

$$\sum_{j=1}^6 \int_{\Omega_e} \{(\nabla \times \mathbf{w}_i) \cdot (\nabla \times \mathbf{w}_j)\} dV [Al]_j + \sum_{j=1}^6 \int_{\Omega_e} i\omega\mu\sigma \mathbf{w}_i \cdot \mathbf{w}_j dV [Al]_j = \mu \int_{\Omega_e} \mathbf{w}_i \cdot \mathbf{i}_{s_e} dV, \dots (7)$$

which is rewritten in the matrix form as,

$$\{[M_e] + i\omega\mu\sigma_e[N_e]\}[\mathbf{A}l_e] = \mathbf{s}_e \dots (8)$$

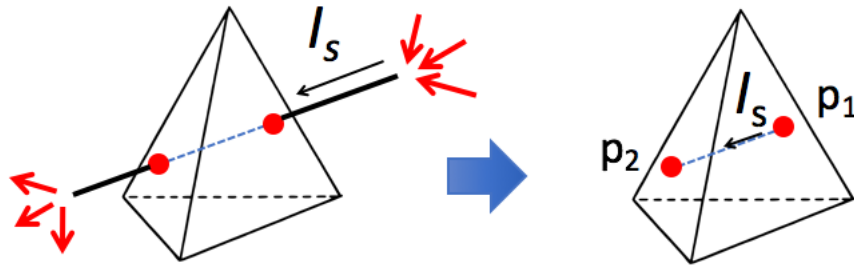
$$[\mathbf{A}l_e] = [Al_1 \ Al_2 \ Al_3 \ Al_4 \ Al_5 \ Al_6]^T \dots (9)$$

$$[M_e]_{ij} = \int_{\Omega_e} \{(\nabla \times \mathbf{w}_i) \cdot (\nabla \times \mathbf{w}_j)\} dV = \frac{1}{3v} \mathbf{x}_{mn} \cdot \frac{1}{3v} \mathbf{x}_{m'n'} \times v = \frac{1}{9v} \mathbf{x}_{mn} \cdot \mathbf{x}_{m'n'} \dots (10)$$

$$\begin{aligned} [N_e]_{ij} &= \int_{\Omega_e} \mathbf{w}_i \cdot \mathbf{w}_j dV = \int_{\Omega_e} (\lambda_k \nabla \lambda_l - \lambda_l \nabla \lambda_k) \cdot (\lambda_{k'} \nabla \lambda_{l'} - \lambda_{l'} \nabla \lambda_{k'}) dV \\ &= \int_{\Omega_e} (\lambda_k \lambda_{k'} \nabla \lambda_l \cdot \nabla \lambda_{l'} - \lambda_k \lambda_{l'} \nabla \lambda_l \cdot \nabla \lambda_{k'} - \lambda_l \lambda_{k'} \nabla \lambda_k \cdot \nabla \lambda_{l'} + \lambda_l \lambda_{l'} \nabla \lambda_k \cdot \nabla \lambda_{k'}) dV \\ &= \nabla \lambda_l \cdot \nabla \lambda_{l'} \int_{\Omega_e} \lambda_k \lambda_{k'} dV - \nabla \lambda_l \cdot \nabla \lambda_{k'} \int_{\Omega_e} \lambda_k \lambda_{l'} dV \\ &\quad - \nabla \lambda_k \cdot \nabla \lambda_{l'} \int_{\Omega_e} \lambda_l \lambda_{k'} dV + \nabla \lambda_k \cdot \nabla \lambda_{k'} \int_{\Omega_e} \lambda_l \lambda_{l'} dV \dots (11) \end{aligned}$$

$$\begin{aligned} s_i^e &= \mu \int_{\Omega_e} \mathbf{w}_i \cdot \mathbf{i}_{s_e} dV = \mu I_s \left(\frac{\mathbf{w}_i^e(r_{p1}) + \mathbf{w}_i^e(r_{p2})}{2} \right) \cdot \overrightarrow{p_1 p_2} \quad (12) \\ &= \frac{\mu I_s (\lambda_k(p_1) \nabla \lambda_l - \lambda_l(p_1) \nabla \lambda_k + \lambda_k(p_2) \nabla \lambda_l - \lambda_l(p_2) \nabla \lambda_k)}{2} \cdot \overrightarrow{p_1 p_2} \\ &= \frac{\mu I_s ((\lambda_k(p_1) + \lambda_k(p_2)) \nabla \lambda_l - (\lambda_l(p_1) + \lambda_l(p_2)) \nabla \lambda_k)}{2} \cdot \overrightarrow{p_1 p_2} \end{aligned}$$

In derivation of Eq. (12), the straight electric current defined by a Heaviside function penetrates the tetrahedron as shown below.



In the derivation of Eq. (9), we used the relationship of

$$\nabla \times \mathbf{w}_j = 2 \nabla \lambda_k \times \nabla \lambda_l = \frac{1}{3v} \mathbf{x}_{mn} \dots (13),$$

Correspondence between (k,l) and (m,n) are listed in Table A1.

In the calculation of Eq. (11), we can use the relationships:

$$\nabla \lambda_k = \frac{1}{6v} \mathbf{x}_{ln} \times \mathbf{x}_{mn} \dots (14)$$

$$\int_{\Omega_e} \lambda_k \lambda_l dV = \begin{cases} 6v \frac{1! 1!}{(1+1+3)!} = \frac{v}{20} & (k \neq l) \\ 6v \frac{2!}{(2+3)!} = \frac{v}{10} & (k = l) \end{cases} \dots (15)$$

Note the theorem for integral of nodal basis function for a tetrahedron:

$$\int_{\Omega_e} \lambda_1^k \lambda_2^l \lambda_3^m \lambda_4^n dV = 6v \frac{k! l! m! n!}{(k+l+m+n+3)!} \dots (16)$$

Appendix B: Scaling of the governing equation for ActFEM calculation

In solving discretized form combining Eq. (7) and (12),

$$\begin{aligned} \sum_{j=1}^6 \int_{\Omega_e} \{(\nabla \times \mathbf{w}_i) \cdot (\nabla \times \mathbf{w}_j)\} dV [Al]_j + \sum_{j=1}^6 \int_{\Omega_e} i\omega\mu\sigma \mathbf{w}_i \cdot \mathbf{w}_j dV [Al]_j \\ = \mu I_s \left(\frac{\mathbf{w}_i^e(r_{p1}) + \mathbf{w}_i^e(r_{p2})}{2} \right) \cdot \overrightarrow{p_1 p_2} \dots (17) \end{aligned}$$

a scaling is implemented in ActFEM. Scaling length L , set to be 1000m in ActFEM, is used. Consider to rewrite Eq. (17) by using $\nabla = \frac{1}{L} \nabla'$, $\mathbf{w} = \frac{1}{L} \mathbf{w}'$, $dV = L^3 dV'$, $\overrightarrow{p_1 p_2} = L \overrightarrow{p_1 p_2}'$, $Al[s \cdot V/m \cdot m] = \int A \cdot d\mathbf{l} = \frac{1}{L} Al'$, where Al' [s · mV/km · km], and $\mathbf{E} = i\omega\mathbf{A}$. These replacements correspond to the fact that $\mathbf{w}$ and ∇ are constructed as $\mathbf{w}'$ and ∇' with the coordinate system in the unit of km.

$$\begin{aligned} \sum_{j=1}^6 \int_{\Omega_e} \left\{ \left(\frac{1}{L} \nabla' \times \frac{1}{L} \mathbf{w}'_i \right) \cdot \left(\frac{1}{L} \nabla' \times \frac{1}{L} \mathbf{w}'_j \right) \right\} L^3 dV' \frac{1}{L} [Al']_j + \sum_{j=1}^6 \int_{\Omega_e} i\omega\mu\sigma \frac{1}{L} \mathbf{w}'_i \cdot \frac{1}{L} \mathbf{w}'_j L^3 dV' \frac{1}{L} [Al']_j \\ = \mu I_s \frac{1}{L} \left(\frac{\mathbf{w}'(r_{p1}) + \mathbf{w}'(r_{p2})}{2} \right) \cdot L \overrightarrow{p_1 p_2}' \\ \Leftrightarrow \sum_{j=1}^6 \int_{\Omega_e} \{(\nabla' \times \mathbf{w}'_i) \cdot (\nabla' \times \mathbf{w}'_j)\} dV' [Al']_j + \sum_{j=1}^6 \int_{\Omega_e} L^2 i\omega\mu\sigma \mathbf{w}'_i \cdot \mathbf{w}'_j dV' [Al']_j \\ = \mu I_s L^2 \left(\frac{\mathbf{w}'(r_{p1}) + \mathbf{w}'(r_{p2})}{2} \right) \cdot \overrightarrow{p_1 p_2}' \end{aligned}$$

Note here that the source current I_s is not scaled, namely in the unit of [A]